

σ . A model is a probability function $p(\cdot)$ over Λ . We say that $p(\cdot)$ is a model of Γ and write $p(\cdot) \models \Gamma$ if and only if $l_i \leq p(S_{\sigma_i}) \leq u_i$ for $i = 1, \dots, k$. Let P_Γ denote the set of all models of Γ .

Initial knowledge is specified by a set of formulas V_0 and two functions $L_0 : V_0 \rightarrow [0, 1]$ and $U_0 : V_0 \rightarrow [0, 1]$. We may then state the following theorem, proved in section D of the supplementary material:

Theorem 2. *Let L_σ and U_σ denote the lower and upper bounds computed by LNN for formula σ . Define $\Gamma_0 = \{(v, L_0(v), U_0(v)) \mid v \in V_0\}$. If $P_{\Gamma_0} \neq \emptyset$, the following inequalities hold:*

$$L_\sigma \leq \inf_{p \in P_{\Gamma_0}} p(S_\sigma) \qquad U_\sigma \geq \sup_{p \in P_{\Gamma_0}} p(S_\sigma)$$

6 Learning

A core strength of the LNN model is its differentiability, permitting the optimization via back-propagation of parameters including operand importance weights, formula truth value bounds, and/or the truth value bounds of atoms. Loss functions for LNN may exploit its logical interpretability, in particular by penalizing contradiction, which can then be used to enforce even complicated logical requirements. An important consideration, however, is whether it is desired to preserve neurons' fidelity to their corresponding logical connectives, especially when presented with classical inputs.

Weighted nonlinear logic behaves classically for classical inputs when optimized as

$$\min_{B, W} \quad E(B, W) + \sum_{k \in N} \max\{0, L_{B, W, k} - U_{B, W, k}\} \quad (6)$$

$$\text{s.t.} \quad \forall k \in N, i \in I_k, \quad \alpha \cdot w_{ik} - \beta_k + 1 \geq \alpha, \quad w_{ik} \geq 0 \quad (6)$$

$$\forall k \in N, \quad \sum_{i \in I_k} (1 - \alpha) \cdot w_{ik} - \beta_k + 1 \leq 1 - \alpha, \quad \beta_k \geq 0 \quad (7)$$

for loss function E , bias vector B , weight matrix W , (disjunction) neuron index set N , and inferred lower and upper bounds $L_{B, W, k}$ and $U_{B, W, k}$ at each neuron. Intuitively, (6) requires disjunctions to return **True** if *any* of their inputs are true, even if their other inputs are 0, i.e. maximally false, while (7) requires them to return **False** if *all* of their inputs are false. Loss function E often embodies typical NN learning objectives such as mean-square error; in addition, *contradiction loss* $\sum_{k \in N} \max\{0, L_{B, W, k} - U_{B, W, k}\}$ penalizes the sum total contradiction observed in the system.

Given the above linear constraints, methods such as Frank–Wolfe [8, 12] may be used to optimize B and W . It is easy to see, however, that weights w_{ik} cannot be made equal to 0, nor can constraints be relaxed to permit nonclassical behavior. This may be corrected via the introduction of slack variables, though the following presents a means of sidestepping this issue while also improving gradients.

6.1 Tailored activation functions

For disjunction with $\beta = 1$, the *tailored activation function* f_w , shown in Figure 1c, is a linear interpolation between four critical points — $(0, 0)$, $(x_F, 1 - \alpha)$, (x_T, α) , and $(x_{\max}, 1)$, establishing regions of unambiguous **True**, intermediate, and **False** truth values, respectively — given

$$f_w(x) = \begin{cases} x \cdot (1 - \alpha) / x_F & \text{if } 0 \leq x \leq x_F, \\ (x - x_F) \cdot (2\alpha - 1) / (x_T - x_F) + 1 - \alpha & \text{if } x_F < x < x_T, \\ (x - x_T) \cdot (1 - \alpha) / (x_{\max} - x_T) + \alpha & \text{if } x_T \leq x \leq x_{\max}, \end{cases} \quad (8)$$

$$x_F = \sum_{i \in I} w_i \cdot (1 - \alpha), \quad x_T = w_{\max} \cdot \alpha, \quad x_{\max} = \sum_{i \in I} w_i.$$

By construction, this guarantees classical inputs produce classical results without the need for constraints. In addition, because x_T is defined in terms of $w_{\max}$, weights may drop to 0 without significantly impacting f_w . By the nature of monotonic linear interpolation, gradients are large and reliable everywhere. Lastly, the tailored activation function establishes α as a means of controlling the system's classicality, with smaller values being more classical.

7 Empirical evaluation

Smokers and friends. LTN experiment $\mathcal{K}_{\text{exp2}}$ [19] has plausible universally quantified axioms for a small universe (a-h) with initial facts for smokes $S(x)$, cancer $C(x)$, and friends $F(x, y)$ (open-world). We repeat the experiment including axioms induced by MLN [15] (total 8 axioms) on this

Table 2: Learnt LNN neuron weights (from P_1^8) and axiom lower bounds (as %) for LTN experiment $\mathcal{K}_{\text{exp2}}$ (universe: a-h) [19], compares LTN degree of satisfiability (as % for 5 axioms) to LNN P_1^5, P_2^5 and repeats in P_1^8, P_2^8 for 8 axioms including 3 induced by MLN [15], with corresponding MLN log-probability weights, followed by axiom-wise contradiction counts. Loss function $(1 + \text{contradiction}) / (1 + \text{factalign} + \text{tightbounds})$ (normalized) component values after training show complete removal of contradictions by relaxing facts and inferences. Gradient descent ($\alpha = 1, \beta = 1, w_{\text{max}} = 1$ with weight normalization and gradient-transparent clamping) adjusts operand weights (P_1), initial axiom/fact bounds (P_1, P_2). Every training epoch performs inference initialized with updated bounds until convergence after the parameter update.

Smokers and friends [$\mathcal{K}_{\text{exp2}}$ (a-h)]	LNN [L, U] 100 epochs, lr: 0.1→0					MLN $L > U$	
	LTN	P_1^5	P_2^5	P_1^8	P_2^8		
$\exists y F(x, y)$	100	100	100	100	100	6.88	0
$\neg F(x, x)$	98	[83, 98]	[83, 98]	[56, 98]	[80, 98]	0.26	0
$\neg F(x, y)^{0.96} \vee F(y, x)^1$	90	[96, 97]	[97, 100]	[51, 95]	[82, 97]	-	0
$\neg S(x)^{0.98} \vee \neg F(x, y)^1 \vee S(y)^{0.97}$	96	[65, 100]	[65, 100]	[65, 100]	[66, 100]	3.53	2
$\neg S(x)^1 \vee C(x)^{0.98}$	77	[57, 100]	[58, 100]	[50, 100]	[60, 100]	-1.35	2
$\neg F(x, y)^{0.97} \vee \neg S(y)^1 \vee F(y, x)^{0.96} \vee S(x)^1$				100	100	6.87	0
$\neg F(w, x)^1 \vee \neg F(w, y)^1 \vee \neg F(z, x)^1 \vee C(z)^{0.97}$				[73, 100]	[70, 100]	4.33	51
$\neg F(w, x)^1 \vee \neg F(y, w)^1 \vee \neg F(z, y)^1 \vee \neg S(y)^{0.99}$				[80, 100]	[77, 100]	9.68	66
Contradiction (remaining)		0	0	0	0		121
Factual $E_i[L_i' - L_i + U_i' - U_i]$ (start: 0.64)		0.42	0.43	0.27	0.37		
Bound tightness $E_i[\exp(L_i - U_i)]$ (start: 1)		0.88	0.89	0.9	0.97		

data, and measure total network contradiction and LNN truth bounds for axioms in Table 2. MLN axiom weight 6.88 for $\exists y F(x, y)$ states that a world with n friendless people is $e^{6.88n}$ times less probable than a world where all have friends, other things being equal [15], but LNN sets full truth as it does not conflict. MLN assigns high log-probability to the last axiom even though it produces contradictions, while LNN sufficiently relaxes the axiom to remove conflicts. LTN assigns high satisfiability of 96 to $\neg S(x) \vee \neg F(x, y) \vee S(y)$, despite evidence against the axiom whereas LNN correctly adjusts bounds to remove contradiction. LNN also infers all logical consequences, in particular friendship symmetry that LTN is unable to produce. Learning of neuron weights can possibly reduce bounds relaxation in P_1^8 when comparing to P_2^8 for the last two high-conflict axioms.

LUBM benchmark. To verify the soundness and completeness of the reasoning performed by LNN, we used Lehigh University Benchmark (LUBM) [10], a synthetic OWL reasoning dataset in university domain with 14 benchmark queries. We generated data for 1 university (102707 triples), parsed the OWL axioms into equivalent LNN constructs resulting in a graph with 257 nodes. After three passes of bidirectional inferences for the network to converge, 14 queries were added to the network to compute respective answers. We compared LNN results with few symbolic reasoners namely Stardog [20], Virtuoso [23] and Blazegraph [2]. Although, all the systems are sound, achieving 100% precision, only Stardog and LNN answers are complete with 100% recall, compared to average recall of 72%(Virtuoso) and 78%(BlazeGraph). In another experiment, we evaluated LNN’s ability to handle ontological inconsistencies, specifically those arising from incorrect axioms by inserting them into network. During inference, those inconsistencies emerge as bound value contradictions, thus allowing LNN to accurately locate and down-weight corresponding nodes from taking part in further inference (section H.2 of the supplementary material expands on noise handling).

TPTP benchmark. We also used a subset of the TPTP benchmark [21, 4] to evaluate LNN in generic classical theorem proving. The TPTP (Thousands of Problems for Theorem Provers) is a comprehensive library for Automated Theorem Proving (ATP) systems, with over 22K problems. We evaluated on a subset of the Common Sense Reasoning domain (937 problems) represented in first-order form (FOF), also filtering out functions and equality — currently not supported by LNN. From the remaining subset of 25 problems, LNN was able to prove all problems within seconds. Despite this small set of TPTP problems, it is noted that none of the recent neural theorem provers (NTP) [6, 17, 5] have demonstrated success on classical ATP. Moreover, NTP inference is limited to

Horn clauses while LNN can support general first-order logical expressions. These promising results show LNN’s potential for generic ATP integrated into an end-to-end differentiable learning system.

8 Conclusions

We have 1) introduced a new conceptual neuro-symbolic framework, including introducing a new class of weighted real-valued logics, and ways of providing truth bounds which we show can have probabilistic semantics, 2) demonstrated approaches for learning in the framework including novel loss function minimizing logical contradiction, and a way to provably bypass the need to perform constrained optimization, and 3) demonstrated approaches for inference/reasoning in the framework, including the upward-downward algorithm which is provably convergent in finite steps, via preliminary experiments confirming the efficacy of the approach compared to others. Planned future work includes approaches for rule induction and mixed symbolic/sub-symbolic sub-networks.

Broader Impact

As a step in the direction of explainable AI, logical neural networks provide a flexible and well-performing framework for neuro-symbolic learning that can nonetheless be 1) interpreted on account of their 1-to-1 correspondence to systems of logical formulae, 2) audited by examining the chain of inferences computed for a given query, and 3) controlled by human users through the specification of logical constraints. As a result, LNNs stand to improve the transparency and fairness of modeled tasks, and may serve as a better performing alternative to older methods (e.g. decision trees and logistic regression) when explainability is a design requirement.

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